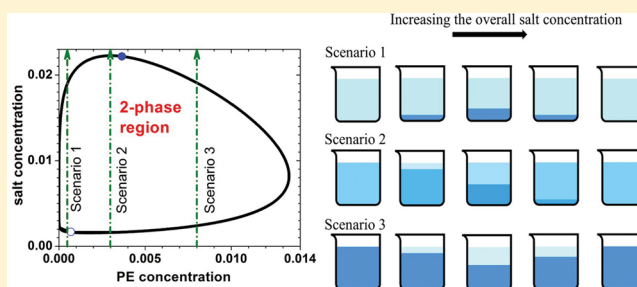


Salting-Out and Salting-In of Polyelectrolyte Solutions: A Liquid-State Theory Study

Pengfei Zhang,[†] Nayef M. Alsaifi,[‡] Jianzhong Wu,[¶] and Zhen-Gang Wang^{*,†,‡}[†]Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, United States[‡]Chemical Engineering Department, King Fahd University of Petroleum & Minerals, Dhahran, Saudi Arabia[¶]Department of Chemical and Environmental Engineering, University of California, Riverside, California 92521, United States

ABSTRACT: We study the phase behavior of polyelectrolyte (PE) solutions with salt using a simple liquid-state (LS) theory. This LS theory accounts for hard-core excluded volume repulsion by the Boublik–Mansoori–Carnahan–Starling–Leland equation of state, electrostatic correlation by the mean-spherical approximation, and chain connectivity by the first-order thermodynamic perturbation theory. We predict a closed-loop binodal curve in the PE concentration–salt concentration phase diagram when the Bjerrum length is smaller than the critical Bjerrum length in salt-free PE solution. The phase-separated region shrinks with decreasing Bjerrum length, and disappears below a critical Bjerrum length. These results are qualitatively consistent with experiments, but cannot be captured by the Voorn–Overbeek theory. On the basis of the closed-loop binodal curve and the lever rule, we predict three scenarios of salting-out and salting-in phenomena with addition of monovalent salt into an initially salt-free PE solution. The Galvani potential—the electric potential difference between the coexisting phases—is found to depend nonmonotonically on the overall salt concentration in some PE concentration range, which is related to the partition of the co-ions in the coexisting phases. Free energy analysis suggests that phase separation is driven by a gain in the electrostatic energy, at the expense of a large translational entropy penalty, due to significant counterion accumulation in the PE-rich phase.



I. INTRODUCTION

Polyelectrolyte (PE) solutions are ubiquitous in biology and in myriads of technological applications, such as cosmetics, wastewater treatment, drug delivery, and gene delivery. The stability of PE solutions, i.e., their thermodynamic phase behavior, is a key property underlying their role in biology and in various applications.¹ For example, in nonviral gene delivery applications,² the negatively charged DNA needs to be condensed into nanoparticles by the addition of polycations. The theoretical understanding of the phase behaviors of PE solutions, however, remains challenging, not only because of the multicomponent nature of the system, but also because of the delicate interplay among various factors, including the translational entropy of each component, excluded volume interactions, chain connectivity, and more importantly the long-range electrostatic interactions. For the simplest salt-free PE solutions, electrostatic interactions can drive a phase separation into a PE-rich phase and a PE-poor phase when the electrostatic interaction strength is sufficiently large; this phenomenon has been studied extensively by various theories^{3–7} and Monte Carlo simulation.⁸ However, for an aqueous PE solution with monovalent counterions, the predicted critical temperature is below the freezing point of water and is thus experimentally inaccessible.⁸

Addition of salt ions can substantially change the phase behavior of PE solutions. Experimentally, it is found that adding

a small amount of salt into a homogeneous salt-free PE solution can induce precipitation of PE chains once the salt concentration is above some threshold value; this is the salting-out phenomenon.^{9–14} Further addition of salt, however, can redissolve the precipitates completely if the salt concentration exceeds a second threshold value; this is the salting-in phenomenon.^{9–14} These salting-out and salting-in phenomena have been observed for both monovalent^{9,13} and multivalent^{10–14} salts in experiments (here “monovalent” or “multivalent” refer to the valency of the counterion of salts); theoretical explanations regarding the mechanism, however, are controversial.^{10,15–20}

For multivalent salts, a popular explanation for the salting-out phenomenon at low salt concentrations^{10,15} is the “bridging” attraction between charged monomers via the adsorbed multivalent counterion due to the local charge inversion at the adsorption site. Phase separation occurs once the strength of this attraction is strong enough to overcome the translational entropy of counterions and the excluded volume repulsion. Further addition of salt will screen this effective attraction and thus lead to salting-in.¹⁰ These arguments, however, do not apply to monovalent salts.^{9,13} For PE solutions

Received: October 5, 2016

Revised: November 23, 2016

Published: December 12, 2016



with monovalent salts, adsorption of a monovalent counterion onto a charged monomer does not induce charge inversion and thus does not generate the “bridging” attraction responsible for salting-out. If salting-out is not caused by the “bridging” attraction, then salting-in at higher salt concentrations cannot be explained by the screening of such attraction.

Thermodynamically, the phase behavior is determined by the free energy of the PE solution. While solvent quality can obviously drive phase separation, here we restrict our consideration to phase separation that is driven purely by electrostatic correlations. In this case, the free energy mainly consists of the translational entropy of each component and the hard-core excluded volume repulsion, in addition to the electrostatic correlations. While electrostatic correlations are the root cause for phase separation, the phase behavior is determined by a delicate balance and interplay among all three free energy contributions, which have different dependences on the concentration of salt ions. The modification in the phase behavior upon addition of salt thus results from the change in the balance among these different contributions.

In this study, we use a simple liquid-state (LS) theory^{4,21} to study the phase behavior of PE solutions with a focus on the effects of salts. We choose this particular LS theory based on the following considerations. First, it affords a closed-form and explicit expression of the Helmholtz free energy in terms of the density of each component in the system, which allows clear identification of the different free energy contributions—the translational entropy of each component, the hard-core excluded-volume repulsion, and electrostatic correlations. Second, this LS theory yields predictions for the thermodynamical properties of PE solutions that compare reasonably well (qualitatively, and, for denser systems, semiquantitatively) with Monte Carlo simulation results.^{8,22} Finally, this LS theory has been used to construct a density-functional theory for the study of inhomogeneous PE solutions.²³

One of the main results of our study is the prediction of a closed-loop binodal curve in the PE concentration vs salt concentration phase diagram at Bjerrum lengths that are smaller than the critical Bjerrum length for the salt-free PE solution. For monovalent co-ions and counterions, the salt co-ion concentration in the PE-poor phase is predicted to be higher than in the PE-rich phase. These results are qualitatively consistent with experiments^{9,13} but cannot be captured by the Voorn–Overbeek theory,^{24,25} where the electrostatic correlation is accounted for by using the Debye–Hückel theory²⁶ for a collection of unconnected point-like charged particles; this suggests the importance of the proper treatment of the electrostatic correlations. In addition, we find that increasing the co-ion valency shrinks the phase-separated region whereas increasing the counterion valency enlarges the phase-separated region. On the basis of the closed-loop binodal curve for PE solutions with monovalent salts, we identify three salting-out and salting-in scenarios with addition of pure salt into an initially salt-free PE solution by applying the lever rule. Furthermore, we find that in general there exists a Galvani potential²⁷—the electric potential difference between the coexisting phases—which is roughly correlated with the ratio of the co-ion concentration between the coexisting phases. Finally, free-energy analysis suggests that phase separation is driven by the gain in electrostatic energy, at the expense of a large translational entropy penalty, as a result of significant counterion accumulation in the PE-rich phase.

II. THEORETICAL FORMULATION

II.A. Liquid-State Theory. Here we briefly describe the theoretical model and the LS theory^{4,21} for PE solutions with salt. For concreteness, we take the PE chains to be negatively charged, i.e., the PEs are polyanions, and the counterions are identical to the small cations from salt. The primitive model²⁶ is used to describe the system, where the solvent is treated as a dielectric continuum with dielectric constant ϵ_r , the small cations (anions) are modeled by positively (negatively) charged hard spheres with diameter σ_+ (σ_-), a polyanion chain is assumed to consist of N tangentially connected negatively charged hard spheres with diameter σ_p . All charges are assumed to be point-like and located at the center of the spheres. Since the solvent is treated implicitly, there are three explicit components in the system: polyanion, cation and anion. Denoting the concentration and the valency of component i as ρ_i and z_i respectively ($i = p, +, -$), global charge neutrality requires that $\sum_i \rho_i z_i = 0$. In order to highlight the effects of electrostatic interactions, we assume no other interactions besides the electrostatic interaction and the hard-core excluded volume repulsion. We write the Helmholtz free energy density as a sum of an ideal part and an excess part, $f = f^{\text{id}} + f^{\text{ex}}$. The ideal part describes the translational degrees of freedom and is known exactly

$$\beta f^{\text{id}} = \frac{\rho_p}{N} \left[\ln \left(\frac{\rho_p}{N} \Lambda_p^3 \right) - 1 \right] + \sum_{i=\pm} \rho_i (\ln \rho_i \Lambda_i^3 - 1) \quad (1)$$

where $\beta \equiv 1/k_B T$ with k_B the Boltzmann constant and T the absolute temperature, Λ_p and Λ_i are length scales arising from integrations over the momentum degrees of freedom, which have no consequence in the phase behavior.

To calculate the excess Helmholtz free energy density, the system is first treated as an ensemble of *disconnected* charged hard spheres, for which the hard-core excluded volume repulsion and the electrostatic correlation are described respectively by the Boublik–Mansoori–Carnahan–Starling–Leland (BMCSL) equation of state^{28,29} and the mean-spherical approximation (MSA).^{30,31} The correction due to chain connectivity is then accounted for by using Wertheim’s first-order thermodynamic perturbation theory (TPT1).^{32–34} The hard-core repulsion of the disconnected charged hard spheres can be accurately described by BMCSL excess free energy density^{28,29}

$$\beta f_{\text{hs}}^{\text{ex}} = -n_0 \ln(1 - \eta) + \frac{n_1 n_2}{1 - \eta} + \frac{n_2^3}{36\pi\eta^3} \left[\eta \ln(1 - \eta) + \frac{\eta^2}{(1 - \eta)^2} \right] \quad (2)$$

with $n_0 \equiv \sum_i \rho_i$, $n_1 \equiv (\sum_i \rho_i \sigma_i)/2$, $n_2 \equiv \pi \sum_i \rho_i \sigma_i^2$, and $\eta \equiv (\pi/6) \sum_i \rho_i \sigma_i^3$. If the diameters for all hard spheres are equal, $\sigma_i = \sigma$, eq 2 reduces to the well-known Carnahan–Starling excess free energy density $\beta f_{\text{hs}}^{\text{ex}} = 6\eta^2(4 - 3\eta)/[\pi\sigma^3(1 - \eta)^2]$.^{35,36}

The electrostatic correlation of the *disconnected* charged hard spheres can be taken into account by using the Ornstein–Zernike equation with the mean spherical approximation (MSA),³⁰ and the relevant excess free energy density is

$$\beta f_{\text{el}}^{\text{ex}} = -l_B \sum_i \frac{\rho_i z_i}{1 + \Gamma \sigma_i} \left[\Gamma z_i + \frac{\pi P_n \sigma_i}{2(1 - \eta)} \right] + \frac{\Gamma^3}{3\pi} \quad (3)$$

where the Bjerrum length $l_B \equiv \beta e^2 / 4\pi\epsilon_0\epsilon_r$, with e the elementary charge and ϵ_0 the vacuum permittivity, characterizes the length scale over which the electrostatic interaction between two unit charges in the solvent is $k_B T$. The screening parameter Γ satisfies

$$\Gamma^2 = \pi l_B \sum_i \frac{\rho_i}{(1 + \Gamma\sigma_i)^2} \left[z_i - \frac{\pi P_n \sigma_i^2}{2(1 - \eta)} \right]^2 \quad (4)$$

and the size asymmetry factor is given by

$$P_n = \sum_i \frac{\rho_i \sigma_i z_i}{1 + \Gamma\sigma_i} \left/ \left[1 + \frac{\pi}{2(1 - \eta)} \sum_i \frac{\rho_i \sigma_i^3}{1 + \Gamma\sigma_i} \right] \right. \quad (5)$$

For the equal-size ($\sigma_i = \sigma$) system, invoking charge neutrality, one gets $P_n = 0$ and eq 3 reduces to $\beta f_{\text{el}}^{\text{ex}} = -\Gamma^3(2/3 + \Gamma\sigma)/\pi$,^{37,38} where Γ satisfies $\Gamma(1 + \Gamma\sigma) = \kappa/2$ with $\kappa \equiv \sqrt{4\pi l_B \sum_i \rho_i z_i^2}$ being the inverse of the Debye screening length.²⁶ In the dilute limit, $\rho_i \rightarrow 0$, $2\Gamma \rightarrow \kappa$ and $\beta f_{\text{el}}^{\text{ex}} \rightarrow -\kappa^3/12\pi$, and thus one recovers the DH prediction of the excess Helmholtz free energy due to electrostatic correlation. While theoretically MSA is often considered a low-order approximation for electrostatic correlations, numerically its predictions are quite good in comparison with MC simulations^{39,40} and in describing simple electrolyte solutions in experiments up to quite high concentrations ($\sim 5\text{M}$) for low and moderate values of l_B .⁴¹ We thus believe that, quantitatively, MSA provides a reasonably good description of the electrostatic correlation energy in a wide range of densities and temperatures. In view of this, while more sophisticated theories, such as the hyper-netted chain approximation³⁶ and DH-extended MSA,⁴² are available to treat the electrostatic correlation, here we choose MSA because it affords a simple, analytical expression.

A simple method to treat the excess Helmholtz free energy due to chain connectivity is the first-order thermodynamic perturbation theory (TPT1), originally proposed by Wertheim for neutral hard sphere chain systems^{32–34} and generalized by Jiang et al. to PE solutions.^{4,21} In TPT1, the chain connectivity contribution is

$$\beta f_{\text{ch}}^{\text{ex}} = \rho_p \left(\frac{1}{N} - 1 \right) \ln \left[g(\sigma_p) \exp \left(\frac{l_B z_p^2}{\sigma_p} \right) \right] \quad (6)$$

where $g(\sigma_p)$ is the contact value of the monomer–monomer radial distribution function of the collection of disconnected charged hard-spheres comprising the polyanion chains. While the MSA provides a simple, analytical form for $g(\sigma_p)$, its combination with the exponential approximation⁴³ gives more accurate predictions in their comparison with Monte Carlo simulation results.^{43,44} The excess Helmholtz free energy due to chain connectivity⁴ is then

$$\beta f_{\text{ch}}^{\text{ex}} = \rho_p \left(\frac{1}{N} - 1 \right) \left\{ \ln \left[\frac{1}{1 - \eta} + \frac{n_2 \sigma_p}{4(1 - \eta)} \right] + \frac{l_B z_p^2}{\sigma_p} - \frac{l_B}{\sigma_p(1 + \Gamma\sigma_p)^2} \left[z_p - \frac{\pi P_n \sigma_p^2}{2(1 - \eta)} \right]^2 \right\} \quad (7)$$

where the sum of the two terms in the first square brackets is the contact value of the monomer–monomer radial distribution function for disconnected neutral hard spheres from the

Percus–Yevick equation,³⁶ and the second and the third terms in the curly brackets account for the electrostatic correlation correction due to chain connectivity. Being a perturbation theory about the state of disconnected units, strictly speaking, TPT1 is only expected to be an accurate description of correlation effects due to chain connectivity in the concentrated regime. However, previous work on salt-free PE solutions has shown that TPT1 yields reasonable predictions for the osmotic coefficient in a wide concentration range, including the dilute and semidilute regimes,^{4,22} as well as the critical temperature and concentration.⁸ We thus believe that the generic effects of the correlations due to chain connectivity are qualitatively captured by the TPT1. While several theories are available to treat the effects of chain connectivity on electrostatic correlations in PE solutions,^{5,6,20} in this work we adopt the TPT1 treatment here for its simplicity, and use it to examine the generic features of the phase diagram due to electrostatic correlations in PE solutions in the presence of salt. A more accurate theoretical treatment of electrostatic correlations per se is not our chief concern in this work.

II.B. Voorn–Overbeek Theory. One of the simplest theories to capture the effects of electrostatic correlations on PE phase behavior is the Voorn–Overbeek (VO) theory.^{24,25} It has been extensively used to study complex coagulation of mixtures of polycation and polyanion solutions.^{24,25,45–47} We include a brief description of the VO theory both because of its widespread use and in order to provide a comparison with our LS theory results. In the VO theory, the free energy density consists of the Flory–Huggins mixing entropy and an electrostatic correlation term, which is taken to be of the Debye–Hückel form²⁶ for an collection of disconnected point-like ions. The hard-core excluded volume interaction is enforced through an incompressibility condition. Chain connectivity is accounted for only in the mixing entropy, but is completely ignored in any excess free energy contributions. In particular, it is ignored in the electrostatic correlation term; thus the VO theory makes no distinction between the small ions and charges on the PE backbone. In the absence of other interactions, the system free energy density f reads

$$\beta f v = \frac{\phi_p}{N} \ln \phi_p + \phi_+ \ln \phi_+ + \phi_- \ln \phi_- + \phi_s \ln \phi_s - \frac{\kappa^3}{12\pi} \quad (8)$$

where ϕ_p , ϕ_+ , ϕ_- , and ϕ_s are the volume fractions of polyanion segment, cation, anion and solvent, respectively; $\phi_s = 1 - \phi_p - \phi_+ - \phi_-$ due to the incompressibility constraint. κ is the inverse of Debye screening length defined in section II.A. In eq 8, it is assumed that the PE segments, small ions, and solvent molecules have the same volume v ; extension is straightforward to allow different volumes for the different species. While there is no direct correspondence between the volume fractions in the VO theory and the densities in the LS theory because of the different treatment of the solvent, the volume fraction of the solutes ϕ_i can crudely be considered to be given by $\phi_i = (\pi/6)\rho_i\sigma_i^3$.

II.C. Construction of Phase Diagram. From the Helmholtz free energy density f , we compute the chemical potential of the species from $\mu_i \equiv \frac{\partial f}{\partial \rho_i} \bigg|_{\rho_{j \neq i}}$. Charge neutrality is

enforced by introducing a Lagrange multiplier Ψ in the grand free energy minimization, which has the interpretation of an electrostatic potential. Phase equilibrium between the PE-poor

phase (denoted by phase I) and its coexisting PE-rich phase (denoted by phase II) is determined by the equality of the electrochemical potential for all components $\mu_i^I + e z_i \Psi^I = \mu_i^{II} + e z_i \Psi^{II}$ and equality of the (osmotic) pressure $P^I = P^{II}$, where $P = \sum_i \rho_i \mu_i - f$.⁴⁸ While there are two Lagrange multipliers Ψ^I and Ψ^{II} for the PE-poor and PE-rich phases, respectively, only their difference is meaningful. We thus define $\Psi_G \equiv \Psi^{II} - \Psi^I$, which can be understood as a Galvani potential.²⁷ Ψ_G , along with the concentration of each component in the coexisting phases, are determined by solving the above equalities under the charge neutrality constraints.

For the salt-free PE solution, there are only two explicit components in the system, e.g., polyanion and its counterion (cation), and thus we have a total of five independent variables, $\rho_p^{I/II}$, $\rho_+^{I/II}$, and Ψ_G , that need to be solved for from the five equations at a given l_B (two for equality of electrochemical potential for the two species, one for pressure equality and two for charge neutrality in each of the coexisting phases). For the PE solution with salt (specified in section II), there are three explicit components in the system—polyanion, cation, and anion (denoted by p , $+$, and $-$, respectively). At a given l_B , for two-phase coexistence, by the Gibbs phase rule, there is one degree of freedom—this is reflected by the fact that there are seven independent variables $\rho_p^{I/II}$, $\rho_+^{I/II}$, $\rho_-^{I/II}$, and Ψ_G but only six equations. Physically, this means that the coexistence forms a one-dimensional binodal curve in the PE concentration–salt concentration phase diagram. For any fixed value of the concentration of one of the components, the values for the other six variables are solved numerically using the Newton–Raphson method⁴⁹ with the residual error in $\max\{|\mu_i^I - \mu_i^{II} + e z_i \Psi_G|, |P^I - P^{II}|, |\sum_i \rho_i^{I/II} z_i|\} < 10^{-12}$. The binodal curve is then constructed by scanning all possible values of the fixed concentration of that component.

To focus on the main issues of interest—the phase behavior and its dependence on salt concentration, we take all the components to have the same diameter σ in this study. Furthermore, we define dimensionless quantities—reduced Bjerrum length $\tilde{l}_B \equiv l_B/\sigma$, reduced concentration $\tilde{\rho}_i \equiv \rho_i \sigma^3$ and reduced Galvani potential $\tilde{\Psi}_G \equiv \beta e \Psi_G$. $\tilde{l}_B$ serves to quantify the strength of electrostatic interaction, which can be adjusted by changing either the temperature T , the dielectric constant ϵ , or the hard-sphere diameter σ . Henceforth, to simplify notation, we will drop the tilde sign $\sim$ in these quantities. Most of our discussions focus on the case of monovalent salt for which we set $z_p = z_- = -z_+ = -1$. We also briefly study the valency effect of the small ions by considering the cases of PE solutions with either divalent co-ions or divalent counterions.

III. RESULTS AND DISCUSSION

III.A. Salt-Free PE Solution. To investigate the salt effect on the phase behavior of PE solutions, it is instructive to first study the salt-free case—an effective one-component system due to the charge neutrality constraint. This system with monovalent counterion has been studied by Jiang et al. using a similar LS theory.⁴ While the translational entropy and hard-core excluded volume repulsion favor the homogeneous phase, the electrostatic correlation can drive a phase separation at large l_B . Figure 1 shows the binodal curves for a salt-free PE solution with monovalent counterion and for a salt-free PE solution with divalent counterion; the PE solution is in a single phase below the binodal and is in two-phase coexistence above it. The tie line, connecting the coexisting compositions in the PE-poor phase and PE-rich phase on the binodal, is horizontal, and

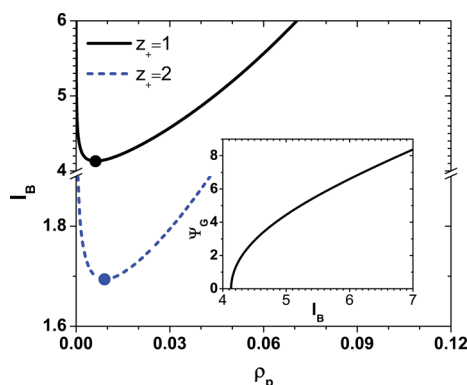


Figure 1. Phase diagram of the salt-free polyelectrolyte (PE) solution. $z_p = -1$ and $N = 100$. The black solid curve is for PE with monovalent counterions and the blue dashed curve is for PE with divalent counterions. Inset: Galvani potential Ψ_G vs Bjerrum length l_B for the salt-free PE solution with monovalent counterions.

shrinks to a point as l_B decreases to the critical $l_{B,c}^{(0)}$. The critical points, specified by the critical concentration $\rho_{p,c}^{(0)}$ and the critical Bjerrum length $l_{B,c}^{(0)}$ are shown as the solid circles in the figure. Close to the critical point, the difference of PE concentration between the two coexisting phases follows $\Delta \rho_p \equiv (\rho_p^{II} - \rho_p^I) \propto (l_B - l_{B,c}^{(0)})^{1/2}$, revealing the mean-field nature of the LS theory. More advanced theory is needed to obtain the correct critical exponents. The system with divalent counterion exhibits similar behavior but with a lower $l_{B,c}^{(0)}$ due to the stronger electrostatic correlation and the smaller translational entropy of the counterions.

Brillantov and co-workers^{50,51} have shown that in dilute salt-free PE solutions, the strong electrostatic interactions at sufficiently low temperatures can induce a condensation of a significant amount of counterions on the chain backbone, which can drive a first-order coil–globule transition. This coil–globule transition will undoubtedly affect the PE concentration in the PE-poor phase, much like how the coil–globule transition affects the phase boundary in neutral polymer solutions under poor solvent conditions.^{52,53} The reported coil–globule transition Bjerrum lengths, however, are much larger than the critical Bjerrum lengths for the macroscopic phase separations examined in our study.

Our predicted Bjerrum length for the salt-free PE solution with monovalent counterions is much higher than the Bjerrum length corresponding to typical experimental aqueous PE solutions at room temperatures;⁵⁴ thus phase separation is not expected for the aqueous salt-free PE solution with monovalent counterions at the room temperature range, and we are not aware of experimental results showing phase separation for such systems. However, for the salt-free PE solution with divalent counterions, our LS theory predicts that phase separation should be accessible for aqueous PE solutions at the room temperature range; thus we believe the predicted phase separation should be observable for such system. We hope our theory will motivate future experiments using PE solutions with divalent counterions.

Because of the intrinsic asymmetry between the polyanion and its counterions in the salt-free PE solution, the electric potential differs between the coexisting PE phases. The electric potential difference Ψ_G between the PE-rich phase and the PE-poor phase—the Galvani potential—increases with the concentration difference between the two coexisting phases, and hence follows similar scaling to $\Delta \rho_p$ near the critical point; this is

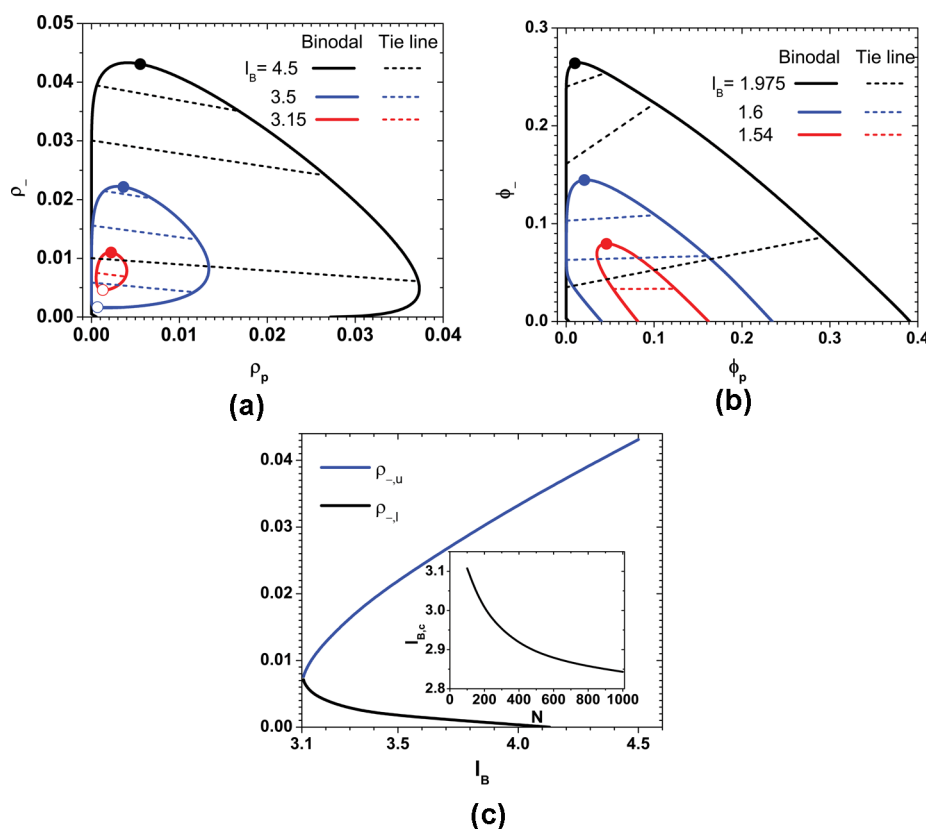


Figure 2. Phase diagram of polyelectrolyte (PE) solution with monovalent salt predicted by (a) the liquid-state (LS) theory and (b) the Voorn-Overbeek theory: solid curve – binodal curve; dashed line – tie line; filled circle – the upper critical point; open circle – the lower critical point. Part c shows the dependence of the upper critical salt concentration (blue curve) and the lower critical salt concentration (black curve) on l_B . The inset shows the chain length dependence of the critical Bjerrum length $l_{B,c}$ for the PE solution with monovalent salts predicted by the LS theory. $N = 100$ is used in parts a and b and the main panel of part c.

shown in the inset of Figure 1 for the case of monovalent counterion.

Previous theories^{4–6} and Monte Carlo simulation⁸ suggest that the critical concentration $\rho_{p,c}^{(0)}$ remains finite and is nearly independent of the chain length when $N > 100$ because of the translational entropy of free counterions, while the critical Bjerrum length $l_{B,c}^{(0)}$ decreases slightly with increasing chain length, due to both the smaller chain translational entropy and the stronger electrostatic correlation. Our results are qualitatively consistent with these expected behaviors (not shown).

III.B. PE Solution with Monovalent Salt. Addition of salt in a PE solution leads to three effects in the system Helmholtz free energy: (1) introduction of the translational entropy of salt ions; (2) increase in the hard-core excluded volume repulsion; and (3) increase in the electrostatic correlation. While the first two effects tend to keep the PE solution in a single, homogeneous state, the last one favors phase separation. These two competing effects are expected to result in nontrivial modifications in the phase diagram from the salt-free PE solution. As we take the polymer to be a polyanion, the concentration of small anions can be used to indicate the salt concentration. In Figure 2a, we show the binodal curves in the polyanion concentration–anion concentration phase diagram for different l_B values. Both the co-ions (anions) and counterions (cations) are taken to be monovalent. The phase-separated region is enclosed by each binodal curve, with the tie lines (indicating the coexisting anion and polyanion concentrations) given by the dashes.

Since the critical Bjerrum length for the salt-free PE solution is at $l_{B,c}^{(0)} \approx 4.13$, it is instructive to distinguish between the $l_B > 4.13$ and $l_B < 4.13$ cases. For $l_B > 4.13$, there is already phase-separation in the salt-free PE solution. As we add salt—moving up from the horizontal axis—the width of the phase-separated region (the length of the tie lines) first increases, i.e., adding salt enhances phase separation. In this low salt concentration regime, the increase in electrostatic correlation due to added salt wins over the increase in translational entropy and hard-core excluded volume repulsion. Further addition of salt, however, narrows the phase-separated window, i.e., shortens the tie lines, until it disappears beyond a certain critical salt concentration; this results from the dominance of the translational entropy and excluded volume increase over the electrostatic correlation gain. Note that the tie lines are sloped downward, indicating that the salt concentration is lower in the PE-rich phase. Because of the finite slope of the tie lines, the critical point at which the length of the tie line vanishes, is not necessarily the highest point in the binodal curve. The critical point is indicated as the black solid circle. We designate this critical point as the upper critical point ($\rho_{p,u}$, ρ_{-u}), to be contrasted with the lower critical point to be discussed below.

For $l_B < l_{B,c}^{(0)}$, the salt-free PE solution is in a single phase at all PE concentrations. However, adding salt can induce phase separation, as shown by the blue and red closed-loop binodal curves in Figure 2(a). At a given l_B (less than $l_{B,c}^{(0)}$) and within certain range of the PE concentration, there exists a threshold value for the salt concentration to cause the system to phase separate, with the coexisting concentrations for the salt anion

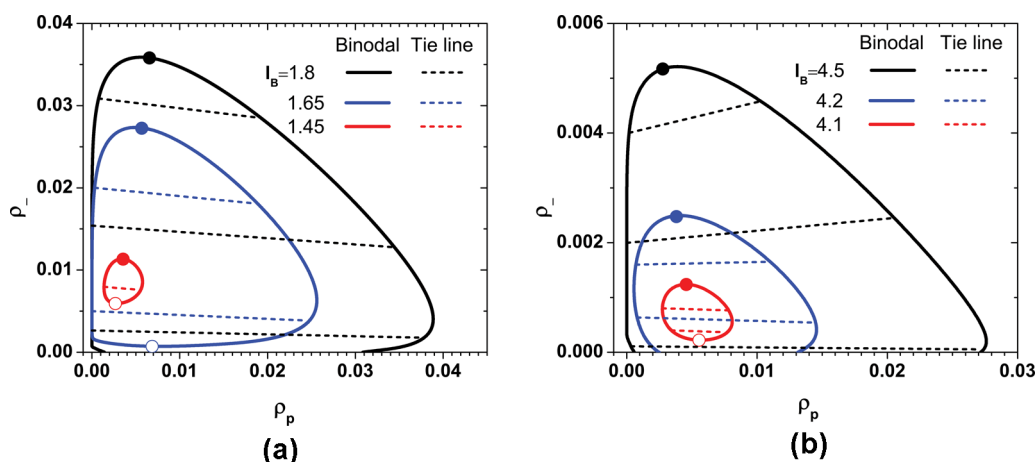


Figure 3. Phase diagram of salty PE solution with (a) divalent counterions (2:1 salt) and (b) divalent co-ions (1:2 salt), predicted by the liquid-state theory, for $N = 100$: solid curve – binodal; dashed line – tie line; filled circles – the upper critical point; open circles – the lower critical point.

and polyanion given by the tie line. Again, the salt anion concentration is slightly higher in the PE-poor phase than its coexisting PE-rich phase. Different from the $l_B > l_{B,c}^{(0)}$ case, there are now two critical points, a lower critical point at $(\rho_{p,l}, \rho_{-,l})$ and an upper critical point at $(\rho_{p,u}, \rho_{-,u})$. The phase-separated region shrinks with decreasing l_B and disappears below another critical Bjerrum length $l_{B,c} \approx 3.11$ —no phase-separation is possible for $l_B < l_{B,c}$. Meanwhile, the lower and the upper critical points approach each other with decreasing l_B and merge at $l_{B,c}$ as shown in Figure 2c. Therefore, for all $l_B < l_{B,c}^{(0)}$, we have a closed-loop binodal curve with an upper critical point and a lower critical point. While the term “upper” and “lower” are defined from the perspective of the salt concentration, we note that the PE concentration at the upper critical point is also higher than the PE concentration at the lower critical point, i.e., $\rho_{p,u} > \rho_{p,l}$. The shape of the closed-loop binodal curve and salt partition behavior in the coexisting phases are qualitatively consistent with experimental data by Eisenberg et al.,⁹ where a similar phase diagram was observed for aqueous solutions of polyvinylsulfonic (PVSA) salts with various monovalent cations, such as mixture of sodium–PVSA with sodium chloride (see Figure 4 in ref 9).

The closed-loop phase diagram is a result of subtle interplay among the different free energy contributions. Of particular importance is the concentration dependence of the electrostatic correlation free energy. To illustrate this point, we have performed the corresponding phase diagram calculations by using the VO theory^{24,25,45,46} for comparison. For the salt-free PE solution, the VO theory predicts a critical Bjerrum length of $l_{B,c}^{(0)} \approx 1.53$. This value is much smaller than the $l_{B,c}^{(0)} \approx 4.13$ predicted by the LS theory, reflecting the overestimate of the electrostatic correlation in VO theory due to its use of the point charge model, relative to the electrostatic correlation in the LS theory. The phase diagram for the salt-free PE solution is qualitatively similar to that in Figure 1 and is not shown. However, qualitatively different from the LS theory result, the VO theory predicts no phase separation for $l_B < 1.53$ when salt is added. Thus, in Figure 2b, we only show the binodal curves for values of l_B larger than 1.53. Closed-loop phase behavior is clearly impossible within the VO theory. Moreover, the VO theory predicts that salt anion concentration is higher in the PE-poor phase than in the PE-rich phase. Furthermore, in the lower part of the phase diagram, with increasing salt concentration at a given overall PE concentration, the PE

concentration in both the PE-poor and PE-rich phases decreases (the overall amount of PE is maintained constant by adjustment in the volume of each phase through the lever rule). These results are in sharp contrast to both the LS theory predictions and the experimental observations.⁹

Although most of our discussions focuses on $N = 100$, the qualitative behavior is unaltered with changing chain lengths and we have also examined the N -dependence in the various properties. As an example, in the inset of Figure 2c, we show the variation of $l_{B,c}$ with N .

III.C. PE Solution with either Divalent Counterion or Divalent Co-Ion. The valency of the salt ions can significantly affect the phase behavior of PE solution. To illustrate this, we consider two cases of salty PE solutions: one with the divalent counterion, and one with the divalent co-ion. Because we take the PE chains to be polyanions, we refer the first system as the PE solution with 2:1 salt and the second system as the PE solution with 1:2 salt.

For the salt-free PE solutions, we have showed in Figure 1 that the critical Bjerrum length $l_{B,c}^{(0)}$ for the divalent counterion system is much lower than that for the monovalent counterion system, due to the much stronger electrostatic correlation and decreased translational entropy of the counterions for the divalent counterion system. This trend persists when 2:1 salts are added. Figure 3a shows that addition of the 2:1 salt results in qualitatively similar phase diagram to the monovalent counterion system presented in Figure 2a, but the phase-separated region for the PE solution with 2:1 salt extends to much lower l_B .

For the system with 1:2 salt, at given PE and co-ion concentrations, more counterions are required to fulfill the charge neutrality constraint than the monovalent co-ion studied earlier. The higher counterion concentration makes it difficult for the system to phase separate due to the larger penalty in its translational entropy. We thus expect that the phase-separated region in the PE solution with 1:2 salt will be smaller than the monovalent co-ion system at the same l_B . Furthermore, we expect the phase-separated region along the l_B axis will be narrower; i.e., the critical Bjerrum length $l_{B,c}$ is larger than the monovalent co-ion system. Both expectations are borne out by the LS theory calculations, as shown in Figure 3b. Moreover, we find that, at larger l_B and higher overall salt concentration $\bar{\rho}_-$, salt co-ions are enriched in the PE-rich phase and depleted in the PE-poor phase (as shown by the tie line of Figure 3a);

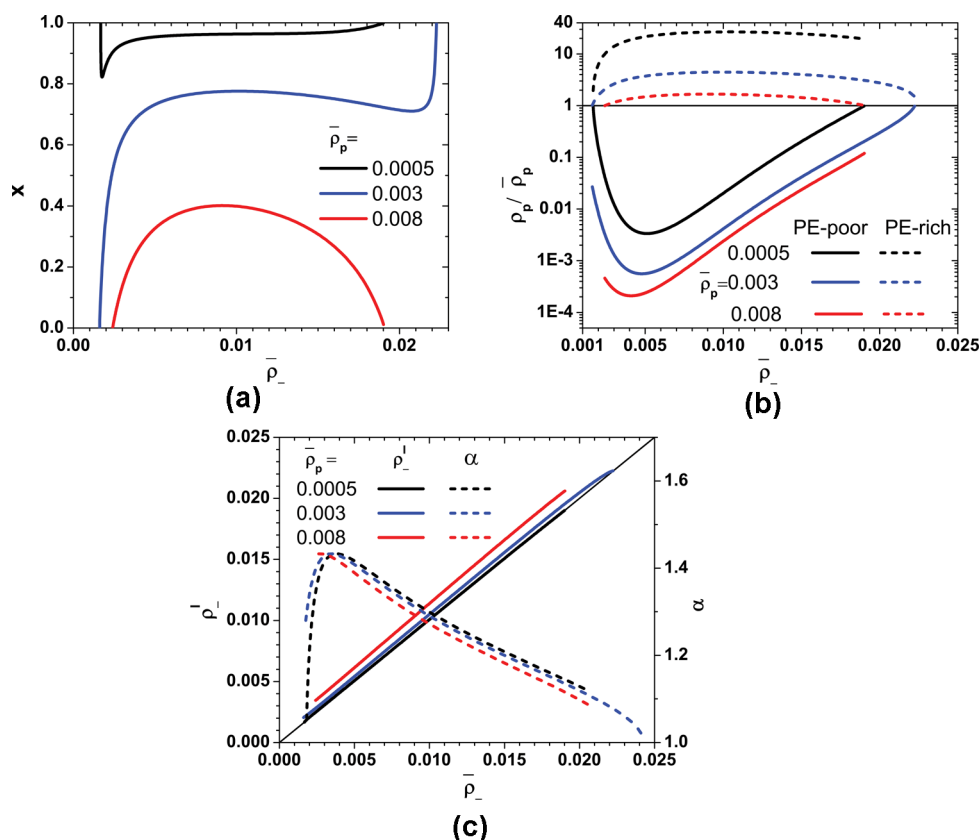


Figure 4. Three scenarios of salting-out and salting-in phenomenon implied by the blue closed-loop binodal in Figure 2a for $N = 100$ and $l_B = 3.5$: (a) the volume fraction of the PE-poor phase (phase I) x vs the overall salt concentration $\bar{\rho}_-$; (b) the ratio of polyanion concentration in the PE-poor phase ρ_p^I and in the PE-rich phase ρ_p^{II} to the overall polyanion concentration $\bar{\rho}_p$; and (c) the anion concentration in the PE-poor phase ρ_-^I and the ratio of the anion concentrations in the PE-poor phase to that in the PE-rich phase $\alpha \equiv \rho_-^I / \rho_-^{II}$.

this behavior is opposite to the monovalent co-ion system studied in Section III.B.

III.D. Implications from the Lever Rule. The closed-loop phase diagram shown in Figure 2a with an upper and a lower critical points, implies interesting salting-out and salting-in behavior in terms of the fractions of the PE-rich and PE-poor phases and the partition of the different components, as a consequence of the lever rule. If we define x to be the volume fraction of the PE-poor phase (phase I), then in the phase-separated region, the lever rule reads $\bar{\rho}_i = x\rho_i^I + (1-x)\rho_i^{II}$, where i refers to either the PE or the co-ion (the counterion concentration is given in terms of the PE and co-ion concentrations by charge neutrality). To illustrate the rich behavior, we take the phase diagram corresponding to $l_B = 3.5$ (the blue binodal in Figure 2a) and examine the phase progression as pure monovalent salt is added to an initially salt-free PE solution (thus maintaining the overall PE concentration constant). We consider only overall PE concentrations $\bar{\rho}_p$ that are bounded by the minimum and maximum PE concentrations on the binodal curve. Depending on the initial overall PE concentration, we predict three different scenarios, depending on the relative location of $\bar{\rho}_p$ to the lower and upper critical points:

Scenario 1: $\bar{\rho}_p < \rho_{p,l}$. Starting from a homogeneous salt-free PE solution, the black curve in Figure 4a shows that addition of salt leads to the emergence of the PE-rich phase (“daughter” phase) when the overall salt concentration reaches the lower phase boundary (salting-out of the PE-rich phase); the volume fraction of this daughter phase $1 - x$ first increases rapidly at

small $\bar{\rho}_-$ and then decreases gradually until it disappears completely at the upper phase boundary (salting-in of the PE-rich phase). These behaviors are shown schematically in the upper panel of Figure 5.

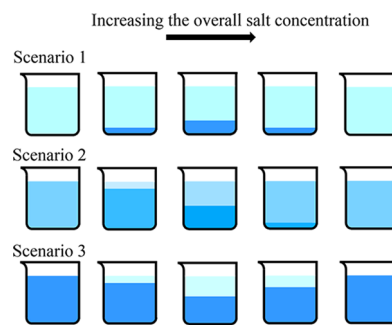


Figure 5. Schematics of the three scenarios of salting-out and salting-in induced by the addition of (pure) monovalent salts into an initially salt-free PE solution, described in Figure 4. The color in the figure is meant to schematically reflect the PE concentrations in each phase but not the small ion concentrations.

Upon phase separation, the concentration of each component in the coexisting phases adjusts to the changing volume fraction of each phase to maintain the phase equilibrium. The black solid and dashed curves in Figure 4b show the evolution of polyanion concentration in the PE-poor phase and its coexisting PE-rich phase with increasing the overall salt concentration $\bar{\rho}_-$. Once $\bar{\rho}_-$ reaches the lower phase

boundary (the salting-out point), the polyanion concentration ρ_p in the PE-poor phase first decreases *continuously* from $\bar{\rho}_p$ while that in the PE-rich phase increases *discontinuously* from $\bar{\rho}_p$. Further addition of salt initially leads to a depletion of the polyanion concentration in the PE-poor phase and an enhancement of the polyanion concentration in the coexisting PE-rich phase, indicating the transfer of PE chains from the PE-poor phase to the emerging PE-rich phase. Upon further increasing $\bar{\rho}_-$, the opposite behavior is obtained, i.e., the polyanion concentration in the PE-poor phase increases and returns to the initial overall PE concentration $\bar{\rho}_p$ at the salting-in point while the polyanion concentration in the PE-rich phase decreases slightly until the PE-rich phase disappears at the salting-in point.

The black solid and dashed curves in Figure 4(c) show, respectively, the evolution of the co-ion concentration in the PE-poor phase (designated as phase I) and the ratio of the co-ion concentration in the PE-poor phase to that in the PE-rich phase (phase II) $\alpha \equiv \rho_-^I/\rho_-^{II}$ for Scenario 1; the diagonal straight line shows the overall co-ion concentration. First, we note that $\alpha > 1$ in the entire phase-separated region, revealing that the co-ion concentration in the PE-poor phase is always higher than in the PE-rich phase, consistent with the negative slope of the tie lines in Figure 2a. Because it is the PE-rich phase that emerges from and disappears into the bulk homogeneous phase with increasing the salt concentration, ρ_- in the PE-poor phase is equal to 1 at both the salting-out and the salting-in points; in between, ρ_- is only slightly larger than $\bar{\rho}_-$ since the PE-poor phase is the majority phase (as shown by the black curve in Figure 4a); ρ_- in the PE-rich phase, however, is discontinuous at both the salting-out point and the salting-in point and is always smaller than $\bar{\rho}_-$ in the phase-separated region (data not shown). Finally, the maximum of α indicates that there exists a specific overall salt concentration at which the co-ion concentrations of the coexisting phases differ the most.

Scenario 2: $\rho_{p,l} < \bar{\rho}_p < \rho_{p,u}$. For Scenario 2, starting from a homogeneous salt-free PE solution, the blue curve in Figure 4a shows that addition of salt leads to the emergence of PE-poor phase (“daughter” phase) when the overall salt concentration reaches the lower phase boundary (salting-out of the PE-poor phase). The volume fraction of this “daughter” phase increases (either monotonically or nonmonotonically, depending on the initial overall PE concentration) until the “mother” phase disappears when $\bar{\rho}_-$ reaches the upper phase boundary (salting-in of the PE-rich phase). We show these behaviors schematically in the middle panel of Figure 5.

The partition of PE chain and salt co-ion in the coexisting phases for Scenario 2, shown by the blue solid and dashed curves in parts b and c of Figure 4, exhibits similar behaviors to those for Scenario 1. However, because it is the PE-poor phase that first emerges upon adding salt, ρ_p and ρ_- in the PE-rich phase are equal to $\bar{\rho}_p$ and $\bar{\rho}_-$ at the salting-out point. Likewise, since it is the PE-rich phase that finally disappears at higher salt concentrations, at the salting-in point, ρ_p and ρ_- in the PE-poor phase return to $\bar{\rho}_p$ and $\bar{\rho}_-$ respectively.

Scenario 3: $\bar{\rho}_p > \rho_{p,u}$. For Scenario 3, starting from a homogeneous salt-free PE solution, the red curve in Figure 4(a) shows that addition of salt leads to the appearance of the PE-poor phase (“daughter” phase) when the overall salt concentration reaches the lower phase boundary (salting-out of PE-poor phase). With further addition of salt, the volume fraction of this “daughter” phase first increases and then

decreases to 0 when $\bar{\rho}_-$ reaches the upper phase boundary (salting-in of the PE-poor phase). A schematic of these salting-out and salting-in behaviors are shown in the lower panel of Figure 5.

The red solid and dashed curves in parts b and c of Figure 4 show that, the partition of PE chain and salt co-ion in the coexisting phases for Scenario 3 exhibits similar behaviors to that for Scenarios 1 and 2, except that, at both the salting-out and the salting-in points, it is in the PE-rich phase rather than in the PE-poor phase that ρ_p and ρ_- are equal to $\bar{\rho}_p$ and $\bar{\rho}_-$ respectively. Finally, we note that the maximum of α shifts leftward with increasing the overall PE concentration, and for higher PE concentration α decreases monotonically as $\bar{\rho}_-$ increases.

The three scenarios of salting-out and salting-in behavior predicted by our work should be readily testable by experiments for aqueous PE solutions with monovalent counterions. Indeed, since the shape of our predicted phase diagram and salt partition (as shown by the blue curve of Figure 2a) agree with the experimental results of ref 9, we may infer that the three scenarios would have been observed in that system. We hope future experiments can specifically test our predictions.

While the above discussions are based on the closed-loop binodal for PE solutions with monovalent salt, similar procedures can be used to predict experimentally interesting salting-out and salting-in phenomenon for PE solutions with divalent co-ion or counterion. For these two systems, the polyanion concentration at the lower critical point can be either smaller or larger than that at the upper critical point; see parts b and c of Figure 3. Qualitatively different behaviors are expected for these two cases. For the case with $\rho_{p,l} < \rho_{p,u}$ we obtain three scenarios of salting-out and salting-in behaviors with addition of pure salt similar to the case for monovalent salt. For the case with $\rho_{p,l} > \rho_{p,u}$ Scenarios 1 and 3 of salting-out and salting-in phenomena are expected to be similar to that for the monovalent salt. For Scenario 2 with $\rho_{p,u} < \bar{\rho}_p < \rho_{p,l}$ however, we expect that it is the PE-rich phase that emerges from the homogeneous bulk phase when the overall salt concentration $\bar{\rho}_-$ exceeds the lower phase boundary (i.e., salting-out of the PE-rich phase), and it is the PE-poor phase that disappears when $\bar{\rho}_-$ reaches the upper phase boundary (i.e., salting-in of the PE-poor phase). The partition of the different components in each phase can be analyzed in a similar way to that for the PE solution with the monovalent salt, and thus will not be discussed here.

III.E. Galvani Potential. The Galvani potential is defined as the electric potential difference between two bulk phases.²⁷ When the PE solution phase separates, the charge asymmetry between the cations and anions in each phase, and the different concentrations between the PE-rich and PE-poor phases, give rise to a potential difference that is generally nonzero. Theoretically, the Galvani potential is related to the Lagrange multiplier that enforces charge neutrality in each phase—it is given by the difference in the value of the Lagrange multiplier between the two phases. Since phase equilibrium requires equality of the *electrochemical* potential for each of the charged species, the Galvani potential is simply the difference in the “chemical” part of the electrochemical potential, which is determined entirely by the concentration. In Figure 6, we show the reduced Galvani potential Ψ_G for the three scenarios corresponding to Figure 4a, with each curve starting from the salting-out point and terminating at the salting-in point. In the

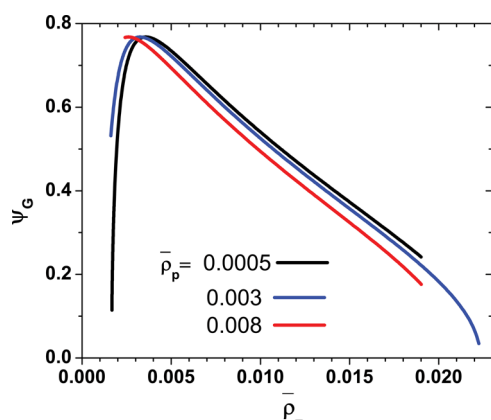


Figure 6. Galvani potential Ψ_G vs the overall salt concentration $\bar{\rho}_-$ for the three scenarios of salting-out and salting-in shown in Figures 4 and 5. $N = 100$.

low $\bar{\rho}_-$ regime, addition of salt enhances phase separation, thus leading to an increase of Ψ_G ; at large $\bar{\rho}_-$, however, further addition of salt diminish the concentration differences between the coexisting phases, thus leading to a decrease of Ψ_G . The maximum of Ψ_G occurs at the $\bar{\rho}_-$ value where the chemical potential difference is the largest. For the small ions, the leading term in the chemical potential is given by the ideal, translational contribution (i.e., $k_B T \ln(\rho_{\pm}^I/\rho_{\pm}^{II})$), thus Ψ_G follows closely the behavior of this concentration ratio. We note that the value of $\bar{\rho}_-$ for the peak in Ψ_G shifts leftward as the PE concentration increases, and for higher PE concentration Ψ_G becomes a strictly monotonically decreasing function of $\bar{\rho}_-$. This can be understood qualitatively by referring to the concentration ratio of co-ions shown in Figure 4c.

III.F. Free Energy Analysis. Thermodynamically, phase separation in a canonical ensemble with fixed overall concentration of each component and total volume takes place when the total Helmholtz free energy of the two coexisting phases is lower than that of the homogeneous phase, or equivalently, $\Delta f \equiv x f^I + (1-x) f^{II} - \bar{f} < 0$. Here $f^{I/II}$ and $\bar{f}$ are, respectively, the Helmholtz free energy densities of Phase I/II with concentrations $\rho_i^{I/II}$, and the homogeneous phase with the overall concentration $\bar{\rho}_i$ for component i . The lever rule $\bar{\rho}_i = x \rho_i^I + (1-x) \rho_i^{II}$ maintains mass conservation for each species. Since $\Delta f = \Delta u - T \Delta s$ where u and s are respectively the internal energy and entropy densities, we examine these terms separately.

In Figure 7a, we show Δf , Δu (in unit of $k_B T$) and Δs (in unit of k_B) as a function of $\bar{\rho}_-$, taking $\bar{\rho}_p = 0.0005$ (Scenario 1) as an example. We see that, in the phase-separated region, both the internal energy density and the entropy density decrease upon phase-separation, but the magnitude of the former is larger than that of the latter. We thus conclude that phase separation is driven by the internal energy gain. While Δu is purely due to electrostatic interactions, Δs is dominated by the change of the translational entropy of cations (counterions) (analysis not shown). Physically, this decrease in the translational entropy corresponds to accumulation of a significant amount of counterions in the PE-rich phase so as to maintain charge neutrality in each phase. The location of the minimum of Δs coincides approximately with that of the largest counterion concentration ratio between the two phases, ρ_+^{II}/ρ_+^I , further confirming the dominance of translational entropy of counterion in the total entropy change. The free energy behaviors for

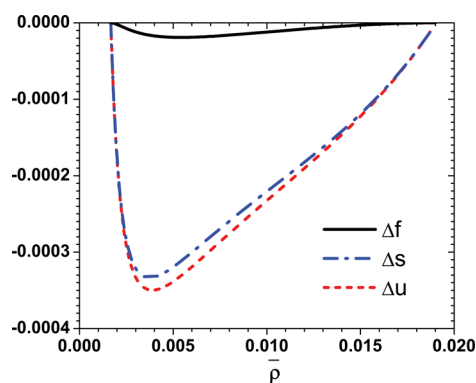


Figure 7. Changes in the Helmholtz free energy, Δf , internal energy Δu and entropy Δs as a function of the overall salt concentration; see the text for the definition of these quantities. $N = 100$ and the overall PE concentration, $\bar{\rho}_p = 0.0005$, are used here.

Scenarios 2 and 3 and their explanations are qualitatively similar to Scenario 1.

IV. SUMMARY AND CONCLUSION

In this work, we have studied the phase behavior of PE solutions both in salt-free condition and with added salt, using a simple liquid-state theory. Our study predicts closed-loop binodal curves in the PE concentration-salt concentration phase diagram at Bjerrum length l_B smaller than the critical value $l_{B,c}^{(0)}$ of salt-free PE solution. The phase-separated region shrinks with decreasing l_B , and disappears when l_B is smaller than another critical Bjerrum length $l_{B,c}$. These results are qualitatively consistent with experiments,⁹ but they cannot be captured by the Voorn–Overbeek (VO) theory,^{24,25} suggesting the importance of the proper treatment of the electrostatic correlations arising from chain connectivity that is missing in the latter. In addition, we find that PE solutions with divalent co-ions exhibit a narrower phase-separated region while PE solutions with divalent counterions have an expanded phase-separated region, relative to the corresponding monovalent systems. On the basis of the closed-loop binodal curve for the PE solution with monovalent salt and the lever rule, we predict three scenarios of salting-out and salting-in phenomenon with addition of pure salt, and calculated the partition of different components in each phase. We have also examined the behavior of the Galvani potential between the coexisting phases and found it to depend nonmonotonically on the overall salt concentration in some PE concentration range, which is related to the partition of the co-ions in the coexisting phases. Finally, free energy analysis shows that phase separation is driven by the gain in electrostatic energy, at the expense of the translational entropy loss due to significant counterion accumulation in the PE-rich phase.

Our present study suggests that the salting-out and salting-in phenomena induced by the addition of salt result from the delicate interplay between the translational entropy, hard-core excluded volume interaction and electrostatic correlation. We note that this understanding differs from the typical arguments made for the multivalent salt, wherein charge-inversion induced bridging attraction at small salt concentration is considered responsible for salting-out, while salting-in at higher salt concentrations is attributed to the screening of such attraction.¹⁰ Thus, we believe that the salting-out/salting-in can be more generally understood in terms of electrostatic

correlations, of which the bridging attraction is a special form for multivalent counterions.

We hope the results presented in this work provide a useful reference for helping guide analyses of experimental data, motivate further experiments and develop improved theories. In this last respect, there are several effects that have not been considered in the current study, for example, the solvent quality effects¹⁹ and counterion condensation.^{55–57} The bulk phase behavior studied here can also be used as a starting point for investigation of inhomogeneous systems, such as the interface between two coexisting phases, and PE solutions under confinements, using density functional theory.²³

AUTHOR INFORMATION

Corresponding Author

*(Z.-G.W.) E-mail: zgw@caltech.edu.

ORCID

Zhen-Gang Wang: 0000-0002-3361-6114

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was conducted jointly by King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia and California Institute of Technology (Caltech) under a collaborative research program in catalysis. The authors gratefully acknowledge the support provided by KFUPM and Caltech. J.W. acknowledges partial financial support from the US National Science Foundation (Grant No. NSF-CBET-820 0852353).

REFERENCES

- (1) Dobrynin, A. V.; Rubinstein, M. Theory of Polyelectrolytes in Solutions and at Surfaces. *Prog. Polym. Sci.* **2005**, *30*, 1049–1118.
- (2) Pack, D. W.; Hoffman, A. S.; Pun, S.; Stayton, P. S. Design and Development of Polymers for Gene Delivery. *Nat. Rev. Drug Discovery* **2005**, *4*, 581–593.
- (3) Mahdi, K. A.; Olvera de la Cruz, M. Phase Diagrams of Salt-Free Polyelectrolyte Semidilute Solutions. *Macromolecules* **2000**, *33*, 7649–7654.
- (4) Jiang, J.; Blum, L.; Bernard, O.; Prausnitz, J. Thermodynamic Properties and Phase Equilibria of Charged Hard Sphere Chain Model for Polyelectrolyte Solutions. *Mol. Phys.* **2001**, *99*, 1121–1128.
- (5) Muthukumar, M. Phase Diagram of Polyelectrolyte Solutions: Weak Polymer Effect. *Macromolecules* **2002**, *35*, 9142–9145.
- (6) Ermoshkin, A.; Olvera de la Cruz, M. A Modified Random Phase Approximation of Polyelectrolyte Solutions. *Macromolecules* **2003**, *36*, 7824–7832.
- (7) Budkov, Yu. A.; Kolesnikov, A. L.; Georgi, N.; Nogovitsyn, E. A.; Kiselev, M. G. A New Equation of State of a Flexible-Chain Polyelectrolyte Solution: Phase Equilibria and Osmotic Pressure in the Salt-Free Case. *J. Chem. Phys.* **2015**, *142*, 174901.
- (8) Orkoulas, G.; Kumar, S. K.; Panagiotopoulos, A. Z. Monte Carlo Study of Coulombic Criticality in Polyelectrolytes. *Phys. Rev. Lett.* **2003**, *90*, 048303.
- (9) Eisenberg, H.; Mohan, G. R. Aqueous Solutions of Polyvinylsulfonic Acid: Phase Separation and Specific Interactions with Ions, Viscosity, Conductance and Potentiometry. *J. Phys. Chem.* **1959**, *63*, 671–680.
- (10) Olvera de la Cruz, M.; Belloni, L.; Delsanti, M.; Dalbiez, J. P.; Spalla, O.; Drifford, M. Precipitation of Highly Charged Polyelectrolyte Solutions in the Presence of Multivalent Salts. *J. Chem. Phys.* **1995**, *103*, 5781–5791.
- (11) Solis, F.; Olvera de la Cruz, M. Collapse of Flexible Polyelectrolytes in Multivalent Salt Solutions. *J. Chem. Phys.* **2000**, *112*, 2030–2035.
- (12) Nguyen, T.; Rouzina, I.; Shklovskii, B. Reentrant Condensation of DNA Induced by Multivalent Counterions. *J. Chem. Phys.* **2000**, *112*, 2562–2568.
- (13) Volk, N.; Vollmer, D.; Schmidt, M.; Oppermann, W.; Huber, K. Conformation and Phase Diagrams of Flexible Polyelectrolytes. *Adv. Polym. Sci.* **2004**, *166*, 29–65.
- (14) Zhang, F.; Skoda, M. W. A.; Jacobs, R. M. J.; Martin, R. A.; Martin, C. M.; Clark, G. F.; Weggler, S.; Hildebrandt, A.; Kohlbacher, O.; Schreiber, F. Reentrant Condensation of Proteins in Solution Induced by Multivalent Counterions. *Phys. Rev. Lett.* **2008**, *101*, 148101.
- (15) Wittmer, J.; Johnner, A.; Joanny, J. Precipitation of Polyelectrolytes in the Presence of Multivalent Salts. *J. Phys. II* **1995**, *5*, 635–654.
- (16) Warren, P. Simplified Mean Field Theory for Polyelectrolyte Phase Behaviour. *J. Phys. II* **1997**, *7*, 343–361.
- (17) Gottschalk, M.; Linse, P.; Piculell, L. Phase Stability of Polyelectrolyte Solutions As Predicted from Lattice Mean-Field Theory. *Macromolecules* **1998**, *31*, 8407–8416.
- (18) Hsiao, P.; Luijten, E. Salt-Induced Collapse and Reexpansion of Highly Charged Flexible Polyelectrolytes. *Phys. Rev. Lett.* **2006**, *97*, 148301.
- (19) Lee, C.-L.; Muthukumar, M. Phase Behavior of Polyelectrolyte Solutions with Salt. *J. Chem. Phys.* **2009**, *130*, 024904.
- (20) Budkov, Yu. A.; Kolesnikov, A. L.; Nogovitsyn, E. A.; Kiselev, M. G. Electrostatic-Interaction-Induced Phase Separation in Solutions of Flexible-Chain Polyelectrolytes. *Polym. Sci., Ser. A* **2014**, *S6*, 697–711.
- (21) Jiang, J.; Liu, H.; Hu, Y.; Prausnitz, J. M. A Molecular-Thermodynamic Model for Polyelectrolyte Solutions. *J. Chem. Phys.* **1998**, *108*, 780–784.
- (22) Chang, R.; Yethiraj, A. Osmotic Pressure of Salt-Free Polyelectrolyte Solutions: A Monte Carlo Simulation Study. *Macromolecules* **2005**, *38*, 607–616.
- (23) Li, Z.; Wu, J. Density Functional Theory for Polyelectrolytes near Oppositely Charged Surfaces. *Phys. Rev. Lett.* **2006**, *96*, 048302.
- (24) Michaeli, I.; Overbeek, J.; Voorn, M. Phase Separation of Polyelectrolyte Solutions. *J. Polym. Sci.* **1957**, *23*, 443–450.
- (25) Overbeek, J.; Voorn, M. Phase Separation in Polyelectrolyte Solutions. Theory of Complex Coacervation. *J. Cell. Comp. Physiol.* **1957**, *49*, 7–26.
- (26) Debye, P.; Hückel, E. The Theory of Electrolytes. I. Lowering of Freezing Point and Related Phenomena. *Phys. Z.* **1923**, *24*, 185–206.
- (27) Mills, I.; Cvitaš, T.; Homann, K.; Kallay, N.; Kuchitsu, K. *Green Book, IUPAC Quantities, Units and Symbols in Physical Chemistry*, 2nd ed.; Blackwell Scientific Publications: Oxford, U.K., 1993; p 59.
- (28) Boublik, T. Hard-Sphere Equation of State. *J. Chem. Phys.* **1970**, *53*, 471–472.
- (29) Mansoori, G. A.; Carnahan, N. F.; Starling, K. E.; Leland, T. W. Equilibrium Thermodynamic Properties of the Mixture of Hard Spheres. *J. Chem. Phys.* **1971**, *54*, 1523–1525.
- (30) Blum, L. Mean Spherical Model for Asymmetric Electrolytes I. Method of Solution. *Mol. Phys.* **1975**, *30*, 1529–1535.
- (31) Hiroike, K. Supplement to Blum's Theory for Asymmetric Electrolytes. *Mol. Phys.* **1977**, *33*, 1195–1198.
- (32) Wertheim, M. Fluids with Highly Directional Attractive Forces. I. Statistical Thermodynamics. *J. Stat. Phys.* **1984**, *35*, 19–34.
- (33) Wertheim, M. Fluids with Highly Directional Attractive Forces. II. Thermodynamic Perturbation Theory and Integral Equations. *J. Stat. Phys.* **1984**, *35*, 35–47.
- (34) Wertheim, M. Thermodynamic Perturbation Theory of Polymerization. *J. Chem. Phys.* **1987**, *87*, 7323–7331.
- (35) Carnahan, N. F.; Starling, K. E. Equation of State for Nonattracting Rigid Spheres. *J. Chem. Phys.* **1969**, *51*, 635–636.
- (36) Hansen, J.-P.; McDonald, I. R. *Theory of Simple Liquids*, 4th ed.; Academic Press: London, 2013.

- (37) Waisman, E.; Lebowitz, J. Mean Spherical Model Integral Equation for Charged Hard Spheres I. Method of Solution. *J. Chem. Phys.* **1972**, *56*, 3086–3093.
- (38) Waisman, E.; Lebowitz, J. Mean Spherical Model Integral Equation for Charged Hard Spheres. II. Results. *J. Chem. Phys.* **1972**, *56*, 3093–3099.
- (39) Abramo, M. C.; Caccamo, C.; Malescio, G.; Pizzimenti, G.; Rogde, S. A. Equilibrium Properties of Charged Hard Spheres of Different Diameters in the Electrolyte Solution Regime: Monte Carlo and Integral Equation Results. *J. Chem. Phys.* **1984**, *80*, 4396–4402.
- (40) Chang, R.; Kim, Y.; Yethiraj, A. Osmotic Pressure of Polyelectrolyte Solutions with Salt: Grand Canonical Monte Carlo Simulation Studies. *Macromolecules* **2015**, *48*, 7370–7377.
- (41) Sanchez-Castro, C.; Blum, L. Explicit Approximation for the Unrestricted Mean Spherical Approximation for Ionic Solutions. *J. Phys. Chem.* **1989**, *93*, 7478–7482.
- (42) Zwanikken, J.; Jha, P.; Olvera de la Cruz, M. A Practical Integral Equation for the Structure and Thermodynamics of Hard Sphere Coulomb Fluids. *J. Chem. Phys.* **2011**, *135*, 064106.
- (43) Anderson, H.; Chandler, D. Optimized Cluster Expansions for Classical Fluids. I. General Theory and Variational Formulations of Mean Spherical Model and Hard Sphere Percus-Yevick Equations. *J. Chem. Phys.* **1972**, *57*, 1918–1929.
- (44) Anderson, H.; Chandler, D.; Weeks, J. Optimized Cluster Expansions for Classical Fluids. III. Applications to Ionic Solutions and Simple Liquids. *J. Chem. Phys.* **1972**, *57*, 2626–2631.
- (45) Spruijt, E.; Westphal, A. H.; Borst, J. W.; Cohen Stuart, M. A.; van der Gucht, J. Binodal Compositions of Polyelectrolyte Complexes. *Macromolecules* **2010**, *43*, 6476–6484.
- (46) van der Gucht, J.; Spruijt, E.; Lemmers, M.; Cohen Stuart, M. A. Polyelectrolyte Complexes: Bulk Phases and Colloidal Systems. *J. Colloid Interface Sci.* **2011**, *361*, 407–422.
- (47) Qin, J.; Priftis, D.; Farina, R.; Perry, S.; Leon, L.; Whitmer, J.; Hoffmann, K.; Tirrell, M.; de Pablo, J. J. Interfacial Tension of Polyelectrolyte Complex Coacervate Phases. *ACS Macro Lett.* **2014**, *3*, 565–568.
- (48) For calculating phase equilibrium in the VO theory, the solvent can be treated either implicitly or explicitly. In the *implicit* solvent treatment, we calculate the electrochemical potential of the i th component of solutes (now considered compressible) and the (osmotic) pressure P from the Helmholtz free energy density. Phase equilibrium at a given temperature is thus determined by the respective equality of the electrochemical potential of each solute component and the pressure in the coexisting phases. In the *explicit* solvent treatment, the system is considered incompressible. The electrochemical potential of each solute component and the chemical potential of the solvent are calculated from the Helmholtz free energy density. Phase equilibrium at a given temperature is determined by the respective equality of the electrochemical potential of each solute component and the solvent chemical potential in the coexisting phases. These two treatments are completely equivalent.
- (49) Press, W. H.; Teukolsky, S. A.; Wetterling, W. T.; Flannery, B. P. *Numerical Recipes: The Art of Scientific Computing*, 3rd ed.; Cambridge University Press: New York, 2007; p 470.
- (50) Brilliantov, N. V.; Kuznetsov, D. V.; Klein, R. Chain Collapse and Counterion Condensation in Dilute Polyelectrolyte Solutions. *Phys. Rev. Lett.* **1998**, *81*, 1433–1436.
- (51) Tom, A. M.; Vemparala, S.; Rajesh, R.; Brilliantov, N. V. Mechanism of Chain Collapse of Strongly Charged Polyelectrolytes. *Phys. Rev. Lett.* **2016**, *117*, 147801.
- (52) Wang, R.; Wang, Z.-G. Theory of Polymers in Poor Solvent: Phase Equilibrium and Nucleation Behavior. *Macromolecules* **2012**, *45*, 6266–6271.
- (53) Wang, R.; Wang, Z.-G. Theory of Polymer Chains in Poor Solvent: Single-Chain Structure, Solution Thermodynamics, and Θ Point. *Macromolecules* **2014**, *47*, 4094–4102.
- (54) Here we assume that the Bjerrum length for the PE solution is the same as that for pure water at room temperature, 7.14 Å, and the hard sphere diameter $\sigma = 4$ Å, from which we can crudely approximate the reduced Bjerrum length as $l_B = 1.785$. However, the dielectric constant of aqueous salt solutions is known to decrease appreciably with salt concentration (see, for example: Wei, Y.-Z.; Chiang, P.; Sridhar, S. Ion Size Effects on the Dynamic and Static Dielectric Properties of Aqueous Alkali Solutions. *J. Chem. Phys.* **1992**, *96*, 4569–4573). Therefore, the effective Bjerrum length for aqueous PE solutions will be larger than 7.14 Å.
- (55) Manning, G. S. Limiting Laws and Counterion Condensation in Polyelectrolyte Solutions. I. Colligative Properties. *J. Chem. Phys.* **1969**, *51*, 924–933.
- (56) Muthukumar, M. Theory of Counter-ion Condensation on Flexible Polyelectrolytes: Adsorption Mechanism. *J. Chem. Phys.* **2004**, *120*, 9343–9350.
- (57) Muthukumar, M.; Hua, J.; Kundagrami, A. Charge Regularization in Phase Separating Polyelectrolyte Solutions. *J. Chem. Phys.* **2010**, *132*, 084901.